Comparison of Data ARC Demand Model and ARC Survey Results

Key Messages

- DHB Shared Services has responses and data from the ARC Quarterly Survey since September 2013.
- Overall, the ARC Quarterly Survey Results shows a higher level of Available Beds/day than Average Residents/day.
- Results for the 2013/14 and 2014/15 Financial Years show that the overall actual supply of Average Beds/day supplied exceeded the forecast Beds/day required for both Demand Model Scenarios.
- Overall results also show for the 2013/14 and 2014/15 Financial Years, there were more Average Residents/day than were forecast by the Demand Model.
- Rest Home Beds increased by 537 Beds between 2013/14 and 2014/15. Rest Home Average Residents decreased by 261 Average Residents per day between the two Financial Years.
- Hospital Beds increased by 773 Beds between 2013/14 and 2014/15. Rest Home Average Residents increased by 321 Average Residents per day over the two financial years.
- Psychogeriatric Beds decreased by 16 Beds between 2013/14 and 2014/15. There was also a decrease in 15 Average Residents per day between the Financial Years.
- Dementia Beds increased by 334 Beds between 2013/14 and 2014/15. There was an increase in Average Residents per day over the two Financial Years of 218 Average Residents

What is ARC Data Talk?

ARC Data Talk is a periodical way to distribute information we consider useful to you about the data that DHB Shared Services receives.

Where is the data sourced from?

This data may come from the Aged Residential Care Bed Survey, the Aged Residential Care Demand Model and a few other sources.

ARC Quarterly Survey Update

Processing of results from the June 2015 Quarter is now complete, and reports to the DHBs have been sent out.

Many thanks to all those organisations that responded, we had a 95% response rate (651 surveys sent out, 617 surveys received at time of cut-off).

ARC Demand Model

Thanks to those of you who supplied feedback on the ARC Demand Model.

We are now working with the developers on the feedback supplied.

If you are interested in attending a workshop that may be offered on the use of the ARC Demand Model, please contact Tim O'Flaherty.

About us

DHB Shared Services is the National operational arm of Technical Advisory Services (TAS).

DHB Shared Services delivers services at a national level, and where appropriate, produces work for All District Health Boards.

TAS' vision is to be an organisation that operates in a business-like fashion, to provide specialised professional services that enable its customers to achieve better health outcomes and value for their money.

TAS' priorities are:

- Do what we currently do well
- Build our reputation
- Deliver new value

For any feedback, or comments, please contact
Tim O'Flaherty
tim.oflaherty@dhbsharedservices.health.nz
DDI: 04 3815154

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Rationale/Background

DHB Shared Services is responsible for coordinating and disseminating ARC information starting with the December 2013 Quarterly Survey, although the first such survey was produced in September 2013.

Information from the ARC Quarterly Survey is processed and checked for accuracy of information before forwarding this information to Health of Older People Portfolio Managers in the DHBs, as well as other Shared Support Agencies in New Zealand.

The ARC Demand Model is a forecasting tool that provides an indication of how many beds may be needed in the service categories of Rest Home Residents, Hospital Residents, Dementia Residents and Psychogeriatric Residents.

The ARC Demand Model is an Excel Macro enabled Workbook that is downloadable free of charge from the DHB Shared Services website. Annual updates of the ARC Demand Model allow for more accurate forecasting, and the Model includes both the Supply and Demand sides for Aged Residential Care. Two main scenarios are used in the ARC Demand Model for forecasting purposes.

- The Population Change scenario is based on estimated 2013/14 per capita utilisation rates (age specific rates) adjusted for population growth projections to 2031
- The Past 5 Year Trend scenario is based on the past 5 years aged care service trends at a national level. These trends are applied to DHB and AREA per capita utilisation rates (age specific) to 2018. From 2019 the projected 2018/19 utilisation rate is adjusted for population growth to 2031.

The ARC Quarterly Survey measures the accuracy and efficacy of the forecast in the Model with the actual supply and demand on a quarterly basis.

Information from the ARC Quarterly Survey is forwarded to Health Partners Consulting Group in order for it to be included in the next release of the ARC Demand Planner.

This ARC Data Talk will compare and contrast differences in where the ARC Demand Model indicated where we would be at the end of the financial years for 2013/14 and 2014/15 with the actuals available from the ARC Quarterly Survey.

Caveats

The information for 2014/15 is based on information on hand while in the closing stages of the June 2015 Quarter.

This information includes data that has been circulated to DHBs in order to verify, and may include pro forma survey responses for facilities prepared after the online survey has been closed. Data may also include proposed corrections to the survey figures that will need to confirmation by District Health Boards.

Such corrections may include corrections to transposed figures and ensuring calculations relating to beds, residents and resident types are consistent. An example of these checks is checking to ensure there are fewer residents than beds available in each facility.

Notes

In the ARC Demand Model, figures for the 2006/07 through to 2013/14 Financial Year for both the Population Change and Past 5 Year Trends are the same, as these figures are a matter of historical record.

As such, the figures under both Population Change and Past 5 Year Trend scenarios for 2013/14 will appear once in the associated tables and charts in this document.

Figures under the Population Change scenario and the Past 5 Year Trend scenario start to diverge from the 2014/15 Financial Year and are considered forecasts. As such there will be two tables presented for the 2014/15 financial year, with figures from each of the scenarios being compared with the actual results from the ARC Survey.

No tests for statistical significance have been done on any of the results included in this document, and ARC Survey Results in charts do not include trend lines.

Average Beds Comparison

2013/14

Differences between the Trend Scenarios and ARC Quarterly Survey

Figure 1 below shows the Population Change and Past 5 Year Trend Scenarios from the ARC Demand Model and Average Beds for 2013/14 from the ARC Quarterly Survey.

Figure 1: Average Bed Comparison per day 2013/14 - ARC Bed Survey v ARC Demand Model

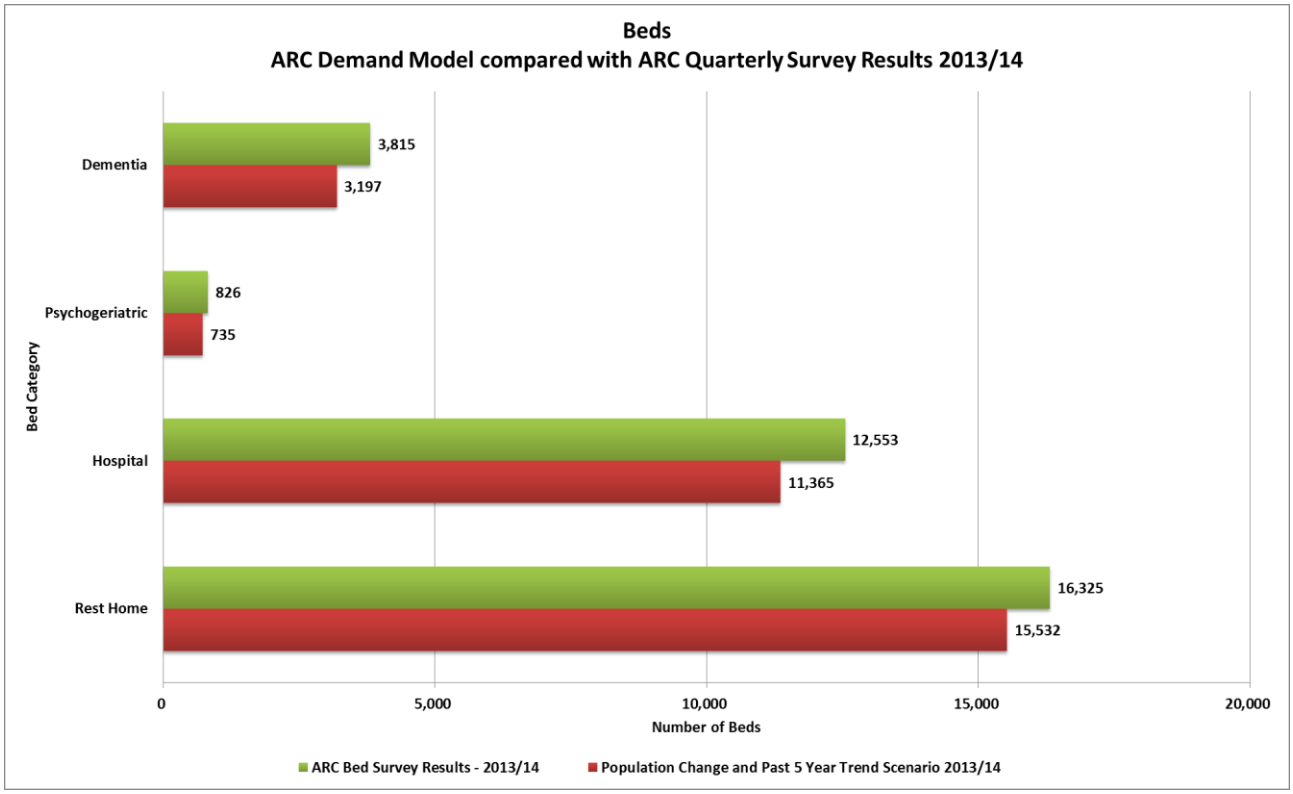


Table 1: Average Beds Comparison per day 2013/14 - ARC Bed Survey v ARC Demand Model

Service Category	ARC Demand Model 2013/14	ARC Quarterly Survey 2013/14	Difference	Accuracy as a Percentage
Rest Home	15,532	16,326	793	95%
Hospital	11,365	12,553	1,188	91%
Psychogeriatric	735	826	91	89%
Dementia	3,197	3,815	618	84%
Totals	30,829	33,520	2,690	92%

Source: ARC Demand Model, ARC Quarterly Survey Results.

Overall, there was an expected demand of 30,829 Beds, matched with an actual Beds supply of 33,520, which is a surplus of 2,690 Beds for the year. As a percentage, the forecast was 92% accurate.

2014/15

From the 2014/15 Financial Year the Population Change and Past 5 Year Trends in the ARC Demand Model start to diverge.

The two tables below show the results of the ARC Demand Model under the Population Change scenario and the ARC Quarterly Results for 2014/15 as well as the results of the ARC Demand Model under the Past 5 Year Trend scenario and the ARC Quarterly Results for 2014/15.

Differences between Population Change Scenario and ARC Quarterly Survey

Table 2: Average Bed Comparison per day 2014/15 - ARC Bed Survey v ARC Demand Model, Population Change Scenario

Service Category	ARC Demand Model Population Change	ARC Survey Result	Difference ARC Quarterly Survey v Population Change	Accuracy as a Percentage
Rest Home	16,126	16,862	736	96%
Hospital	11,813	13,326	1,513	89%
Psychogeriatric	756	810	54	93%
Dementia	3,304	4,149	845	80%
Totals	31,998	35,147	3,149	91%

Source: ARC Demand Model, ARC Quarterly Survey Results.

Under the Population Change Scenario for the 2014/15 Financial Year, the ARC Demand Model indicated an expectation of 31,998 Beds, while there was an actual supply of 35,147 Beds on average for any day in the Financial Year. The difference between the forecast (demand model) and the actual (ARC Survey) was an increase of 3,149 Beds on average for any day. Comparison of the forecast and actual demands shows the forecast was 91% accurate.

Differences between Past 5 Year Scenario and ARC Quarterly Survey

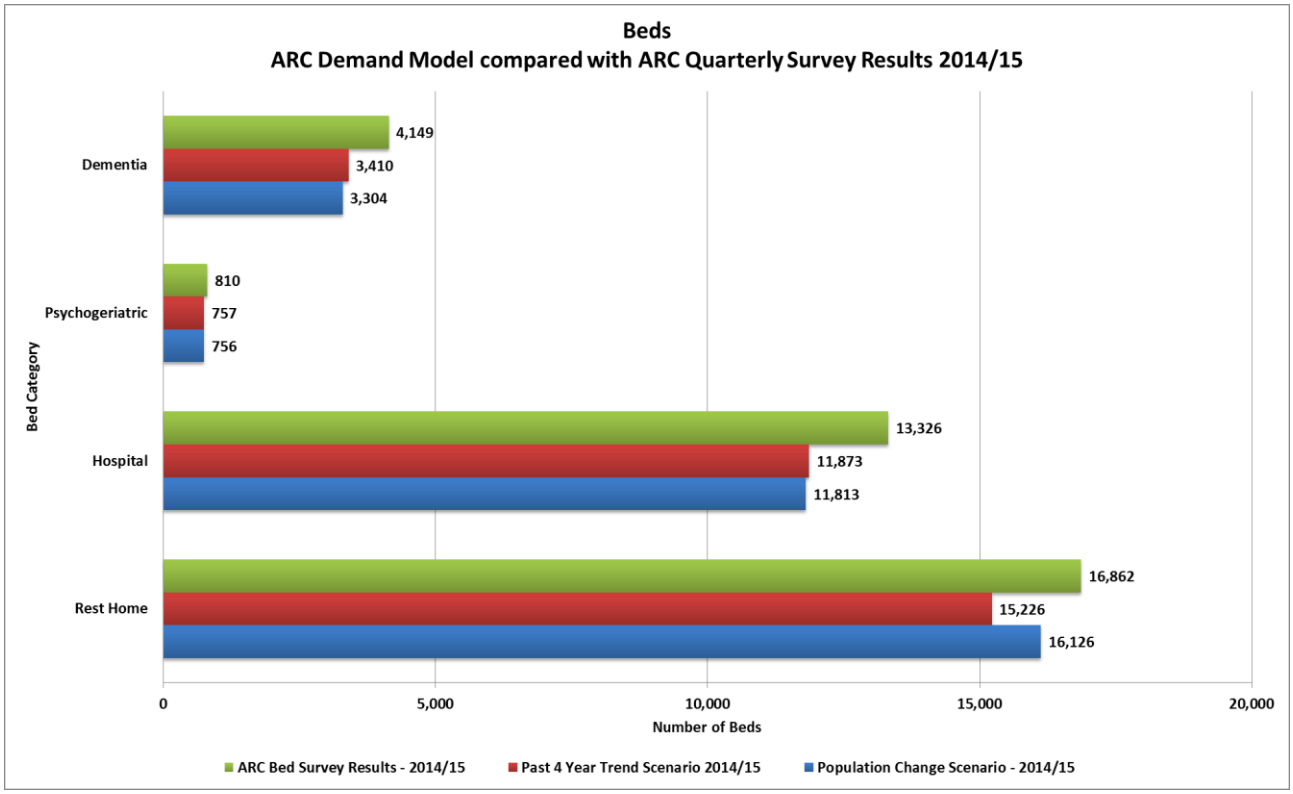
Table 3: Average Bed Comparison per day 2014/15 - ARC Bed Survey v ARC Demand Model, Population Change Scenario

Service Category	ARC Demand Model Past 5 Year Trend	ARC Survey Result	Difference ARC Quarterly Survey v Population Change	Accuracy as a Percentage
Rest Home	15,226	16,862	1,636	90%
Hospital	11,873	13,326	1,453	89%
Psychogeriatric	757	810	53	93%
Dementia	3,410	4,149	739	82%
Totals	31,266	35,147	3,881	89%

Source: ARC Demand Model, ARC Quarterly Survey Results.

Under the Past 5 Year Change Scenario for the 2014/15 Financial Year, the ARC Demand Model indicated an expectation of 31,266 Total Beds, while there was an actual supply of 35,147 Beds on average for any day in the Financial Year. The difference between the forecast (demand model) and the actual (ARC Survey) was an increase of 3,881 Beds on average for any day. Comparison of the forecast and actual demands shows the forecast was 89% accurate.

Figure 2: Average Bed Comparison per day 2014/15 - ARC Bed Survey v ARC Demand Model



Average Residents Comparison

2013/14

Differences between the Trend Scenarios and ARC Quarterly Survey

Table 4: Average Resident Comparison per day 2013/14 - ARC Bed Survey v ARC Demand Model

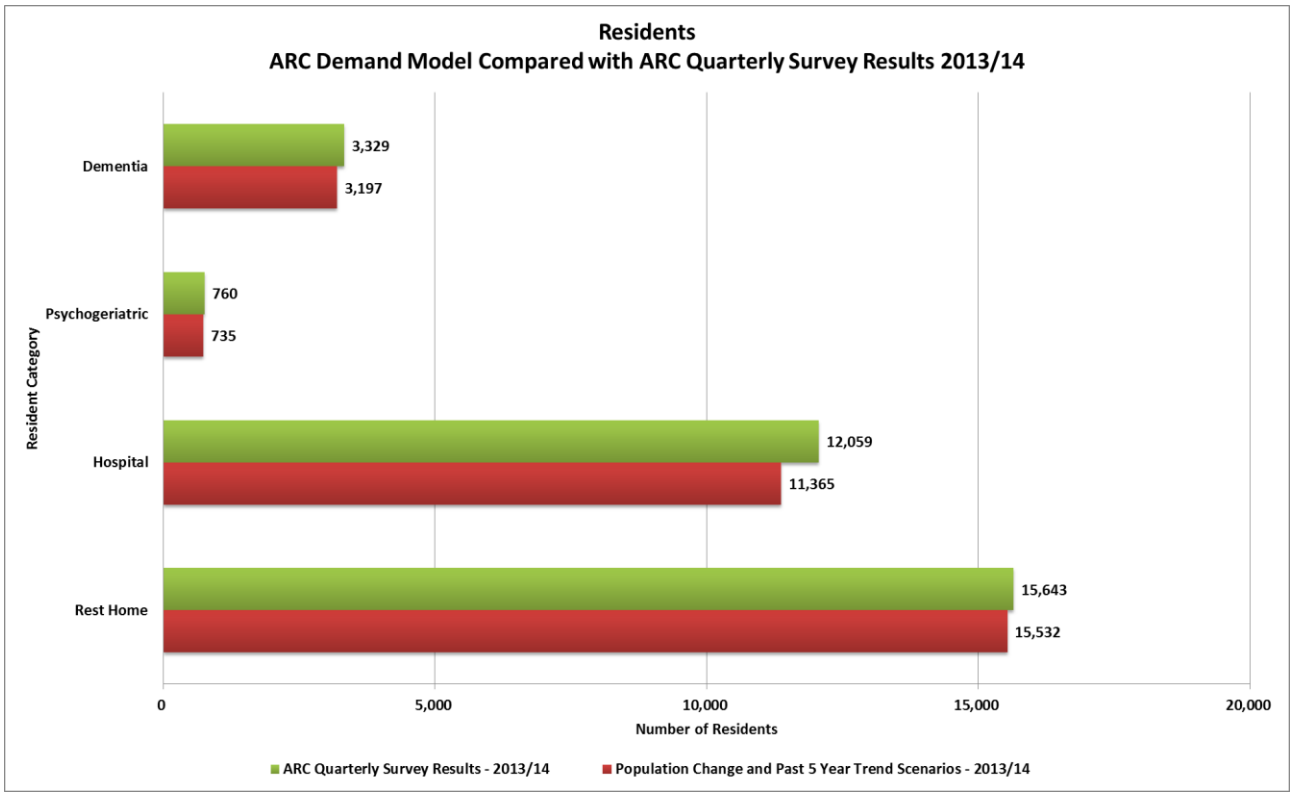
Service Category	ARC Demand Model 2013/14	ARC Quarterly Survey 2013/14	Difference	Accuracy as a Percentage
Rest Home	15,532	15,643	111	99%
Hospital	11,365	12,059	694	94%
Psychogeriatric	735	760	25	97%
Dementia	3,197	3,329	132	96%
Totals	30,829	31,791	962	97%

Source: ARC Demand Model, ARC Quarterly Survey Results.

On average for any day in the 2013/14 Financial Year, there was an expected 30,829 Average Residents, and an actual number of 31,791 Average Residents. The difference between the forecast (demand model) and the actual (ARC Survey) is therefore 962 Average Residents on any given day. Comparing the Forecast and Actual supply shows that the forecast figure was 97% accurate.

Figure 3 below shows the Population Change and Past 5 Year Trend Scenarios from the ARC Demand Model and actual Residents for 2013/14 from the ARC Quarterly Survey.

Figure 3: Average Resident Comparison per day 2013/14 - ARC Bed Survey v ARC Demand Model



2014/15

The tables below show the results of the ARC Demand Model under the Population Change scenario and the ARC Quarterly Results for 2014/15 as well as the results of the ARC Demand Model under the Past 5 Year Trend scenario and the ARC Quarterly Results for 2014/15.

Differences between Population Change Scenario and ARC Quarterly Survey

Table 5: Average Resident Comparison per day 2014/15 - ARC Bed Survey v ARC Demand Model, Population Change Scenario

Service Category	ARC Demand Model Population Change	ARC Survey Result	Difference ARC Quarterly Survey v Population Change	Accuracy as a Percentage
Rest Home	16,126	15,382	-744	95%
Hospital	11,813	12,380	567	95%
Psychogeriatric	756	745	-11	99%
Dementia	3,304	3,547	243	93%
Totals	31,999	32,054	55	100%

Source: ARC Demand Model, ARC Quarterly Survey Results.

For the 2014/15 Financial Year, the ARC Population Change Demand Model indicated an expectation of 31,999 Average Residents per day. There was an actual demand for 32,054 Average Residents per day. The difference between the forecast (demand model) and the actual (ARC Survey) was 55 Average Residents for any day. Comparing the Forecast and Actual supply shows that the forecast figure was 100% accurate.

Differences between Past 5 Year Scenario and ARC Quarterly Survey

Table 6: Average Resident Comparison per day 2014/15 - ARC Bed Survey v ARC Demand Model, Past 5 Year Trend Scenario

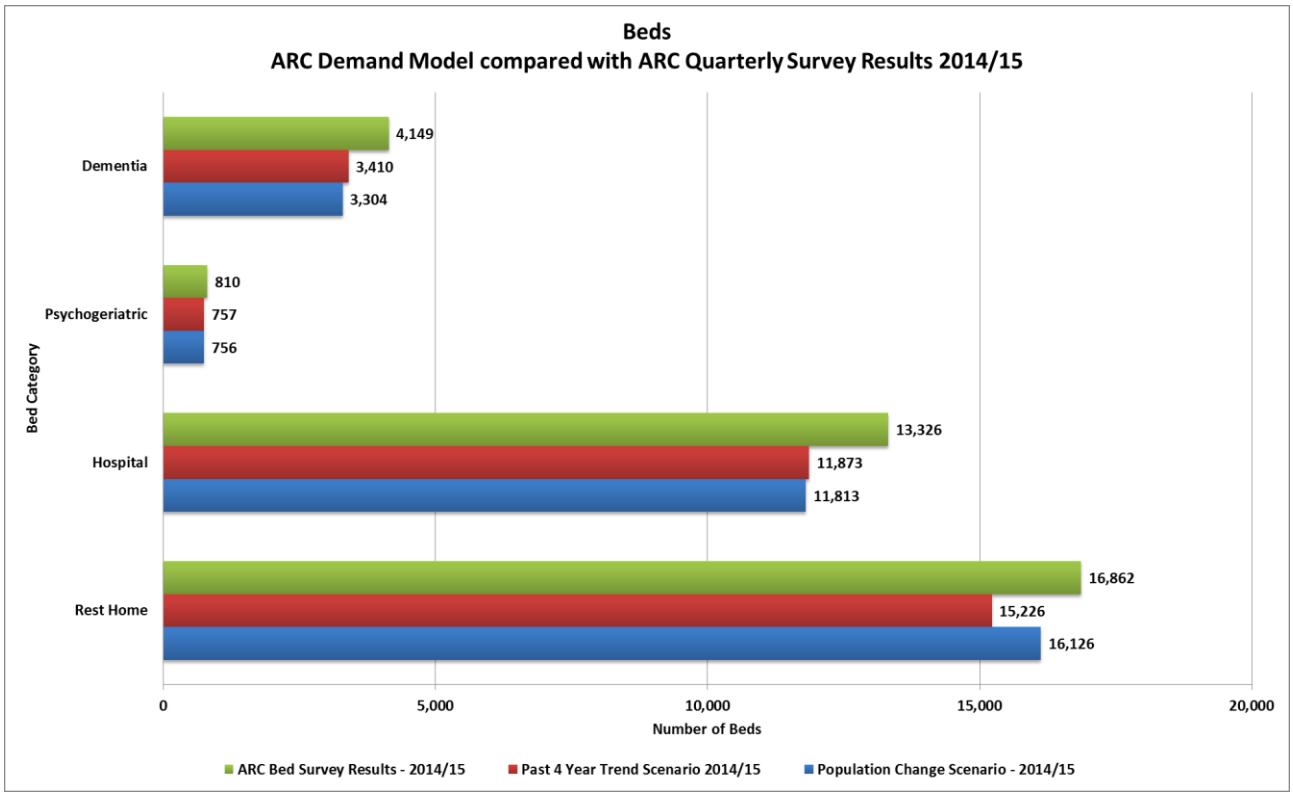
Service Category	ARC Demand Model Past 5 Year Trends	ARC Survey Result	Difference ARC Quarterly Survey v Past 5 Year Trend	Accuracy as a Percentage
Rest Home	15,226	15,226	156	99%
Hospital	11,873	12,380	507	96%
Psychogeriatric	757	745	-12	98%
Dementia	3,410	3547	137	96%
Totals	31,266	32,054	789	98%

Source: ARC Demand Model, ARC Quarterly Survey Results.

For the 2014/15 Financial Year, the ARC Demand Model indicated an expectation of 31,266 Average Residents per day. There was an actual demand for 32,054 Average Residents for any day in the Financial Year. The difference between the forecast (demand model) and the actual (ARC Survey) was 789 Residents on average for any day. The forecast was therefore 98% accurate.

Figure 4 below shows the results of the Population Change and Past 5 Year Trend Scenarios from the ARC Demand Model and actual Bed Days for 2014/15 from the ARC Quarterly Survey.

Figure 4: Average Resident Comparison per day 2014/15 - ARC Bed Survey v ARC Demand Model



Comparison of ARC Survey Results

Beds 2013/14 and 2014/15

The following table details the numbers at the end of each financial year as well as the difference between the two full years of survey information.

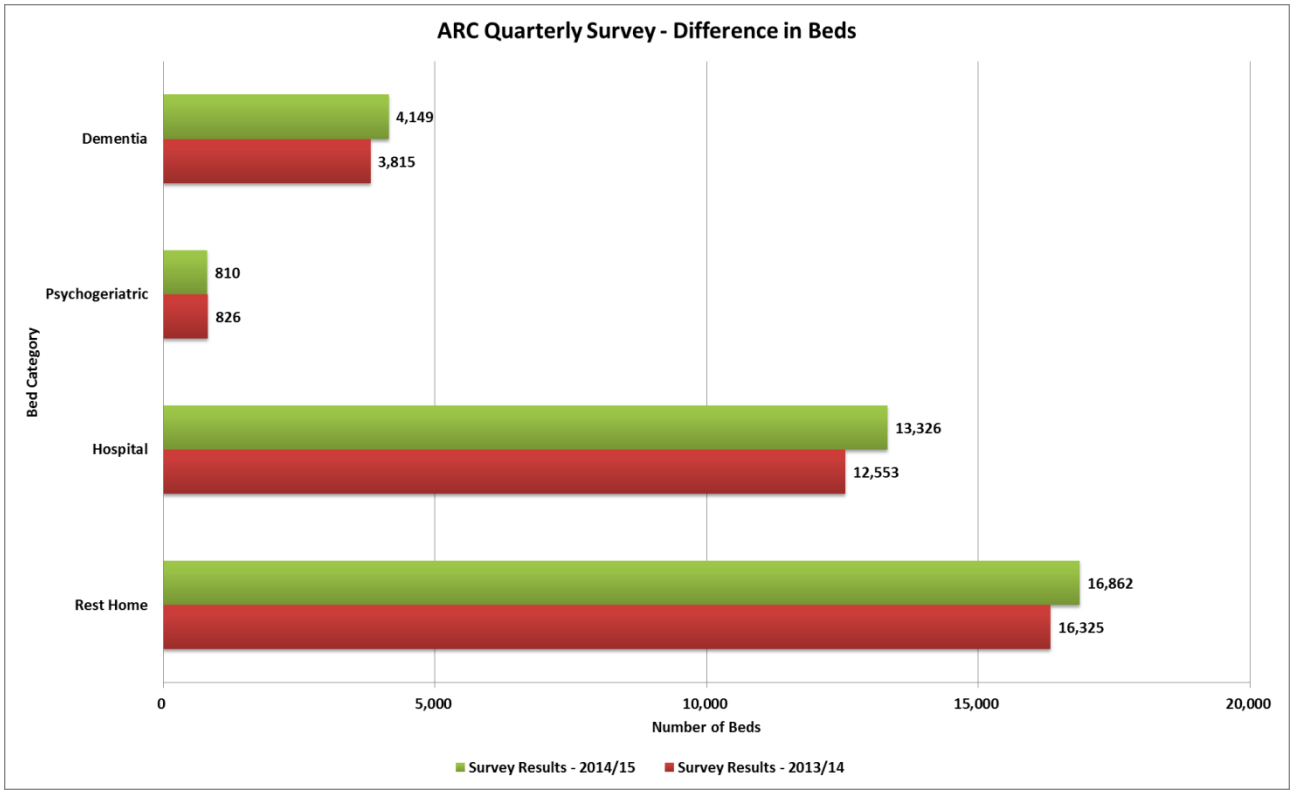
Table 7: Aged Residential Care Quarterly Survey (Bed Days) results

Service Category	ARC Quarterly Survey Beds 2013/14	ARC Quarterly Survey Beds 2014/15	Difference in surveys	Difference as a Percentage
Rest Home	16,325	16, 862	537	3.3%
Hospital	12,553	13,326	773	5.8%
Psychogeriatric	826	810	-16	-2.0%
Dementia	3,815	4,149	334	8.1%
Totals	33,519	35,147	1,628	4.6%

Source: ARC Quarterly Survey Results.

For all service categories, there was an increase from 33,519 Beds to 35,147 Beds, or a total increase of 1,628 Beds, which was a 4.9% increase between the two Financial Years.

Figure 5: ARC Survey Bed Results over two financial years



Average Residents 2013/14 and 2014/15

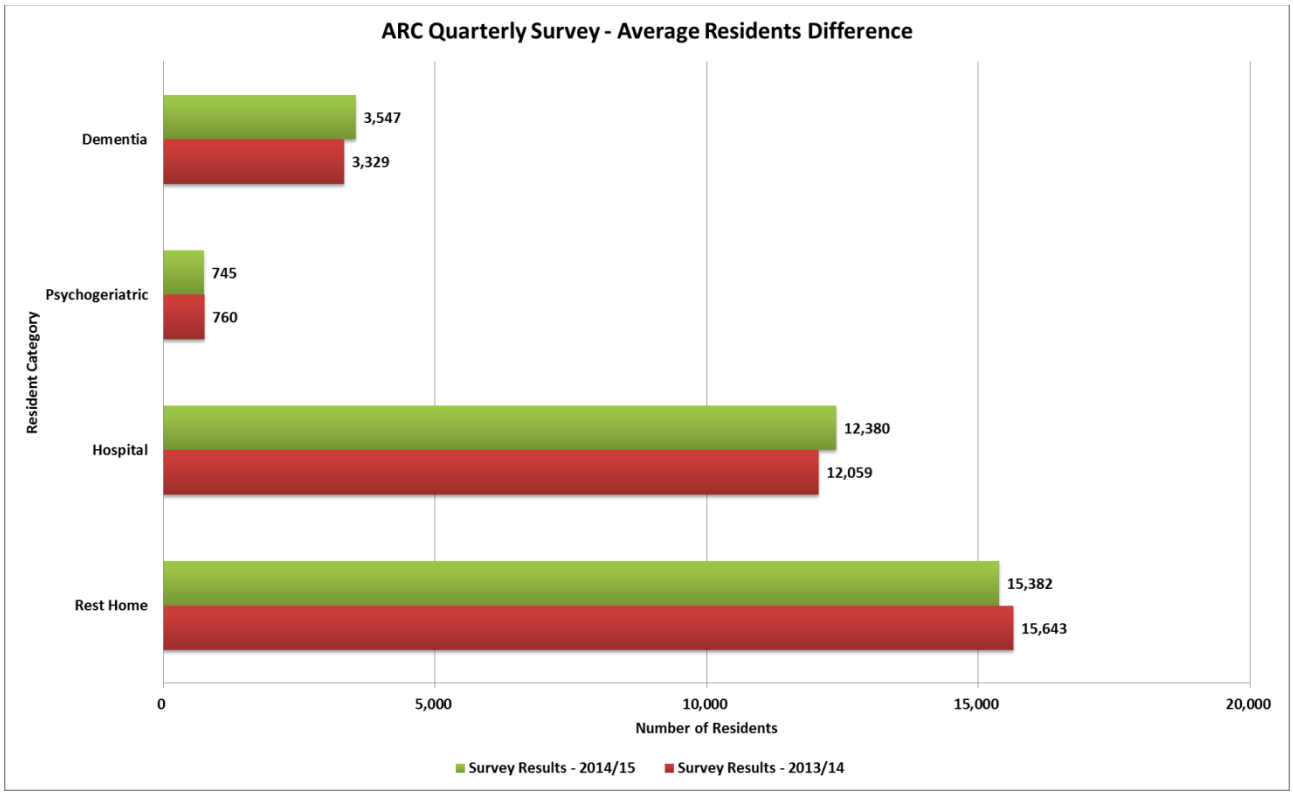
Table 8: Aged Residential Care Quarterly Survey (Average Residents) results

Service Category	ARC Quarterly Survey Beds 2013/14	ARC Quarterly Survey Beds 2014/15	Difference in surveys	Difference as a Percentage
Rest Home	15,643	15,382	-261	-1.7%
Hospital	12,059	12,380	321	2.6%
Psychogeriatric	760	745	-15	-2.0%
Dementia	3,329	3,547	218	6.1%
Totals	31,791	32,054	263	0.8%

Source: ARC Quarterly Survey Results.

For all service categories, there was an increase from 31,791 Average Residents/day to 32,054 Average Residents/day. This was an increase of 263 Average Residents/day, or 0.8% increase between the two Financial Years.

Figure 6: ARC Survey Residents Results over two financial years



Actuals v Projections

The following table and charts display the results of the ARC Demand Model and the ARC Quarterly Survey, under both model scenarios using both the Bed Days and Average Residents measures.

Please note that trend lines do not appear on the charts, due to insufficient data points at the end of financial years for the ARC Results

Table 9: Service Categories, Bed Days measure for ARC Quarterly Survey v ARC Demand Model

		2013/14	2014/15	2015/16	2016/17	2017/18	2018/19	2019/20	2020/21	2021/22
Rest Home (Bed Totals)	Population Change	15,532	16,126	16,686	17,272	17,887	18,531	19,201	19,901	20,630
	Past 5 Year Trend	15,532	15,226	14,888	14,576	14,286	14,020	14,528	15,059	15,613
	ARC Survey Result	16,325	16,862							
Hospital (Bed Totals)	Population Change	11,365	11,813	12,237	12,682	13,148	13,636	14,141	14,668	15,219
	Past 5 Year Trend	11,365	11,873	12,365	12,886	13,438	14,022	14,543	15,088	15,656
	ARC Survey Result	12,553	13,326							
Psychogeriatric (Bed Totals)	Population Change	735	756	780	805	830	857	886	917	949
	Past 5 Year Trend	735	757	782	809	837	868	898	929	962
	ARC Survey Result	826	810							
Dementia (Bed Totals)	Population Change	3,197	3,304	3,416	3,533	3,655	3,783	3,923	4,069	4,222
	Past 5 Year Trend	3,197	3,410	3,640	3,889	4,158	4,448	4,614	4,786	4,966
	ARC Survey Result	3,815	4,149							

Source: ARC Demand Model, ARC Quarterly Survey Results.

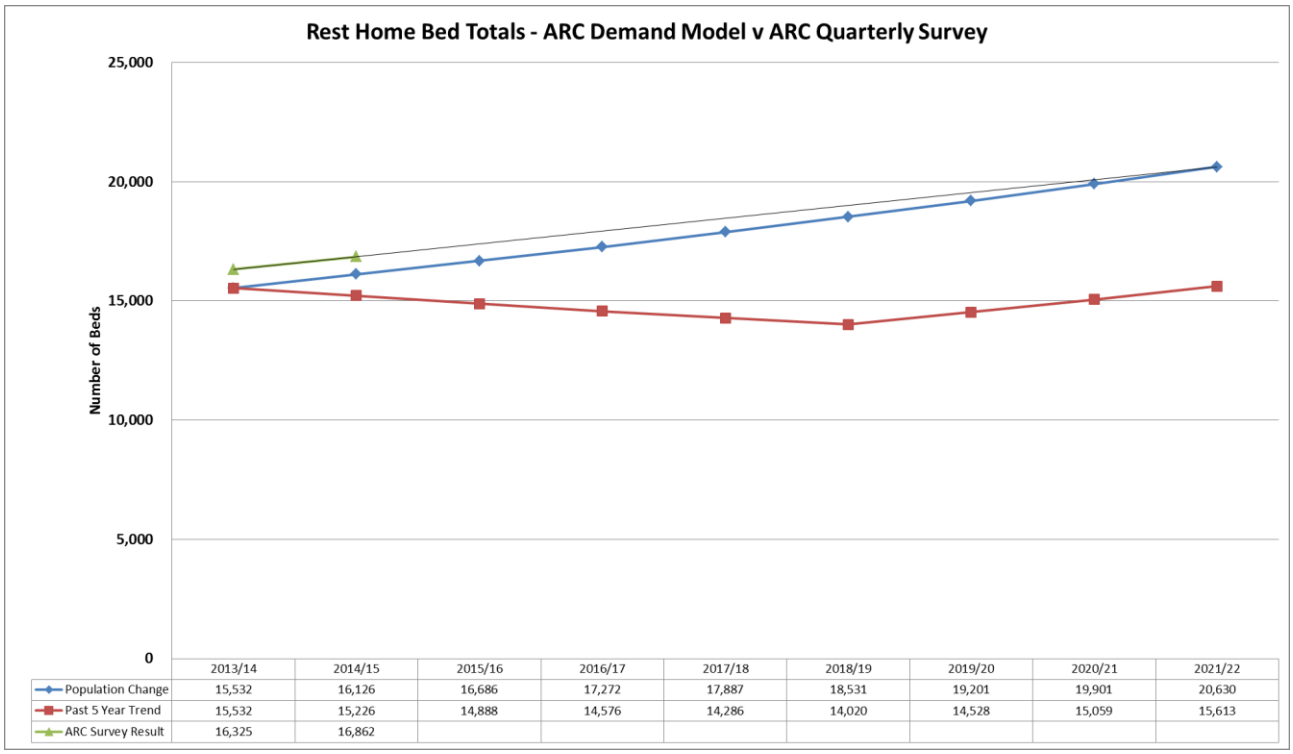
Table 10: Service Categories, Average Residents measure for ARC Quarterly Survey v ARC Demand Model

		2013/14	2014/15	2015/16	2016/17	2017/18	2018/19	2019/20	2020/21	2021/22
Rest Home (Average Residents)	Population Change	15,532	16,126	16,686	17,272	17,887	18,531	19,201	19,901	20,630
	Past 5 Year Trend	15,532	15,226	14,888	14,576	14,286	14,020	14,528	15,059	15,613
	ARC Survey Result	15,643	15,382							
Hospital (Average Residents)	Population Change	11,365	11,813	12,237	12,682	13,148	13,636	14,141	14,668	15,219
	Past 5 Year Trend	11,365	11,873	12,365	12,886	13,438	14,022	14,543	15,088	15,656
	ARC Survey Result	12,059	12,380							
Psychogeriatric (Average Residents)	Population Change	735	756	780	805	830	857	886	917	949
	Past 5 Year Trend	735	757	782	809	837	868	898	929	962
	ARC Survey Result	760	745							
Dementia (Average Residents)	Population Change	3,197	3,304	3,416	3,533	3,655	3,783	3,923	4,069	4,222
	Past 5 Year Trend	3,197	3,410	3,640	3,889	4,158	4,448	4,614	4,786	4,966
	ARC Survey Result	3,329	3,547							

Source: ARC Demand Model, ARC Quarterly Survey Results.

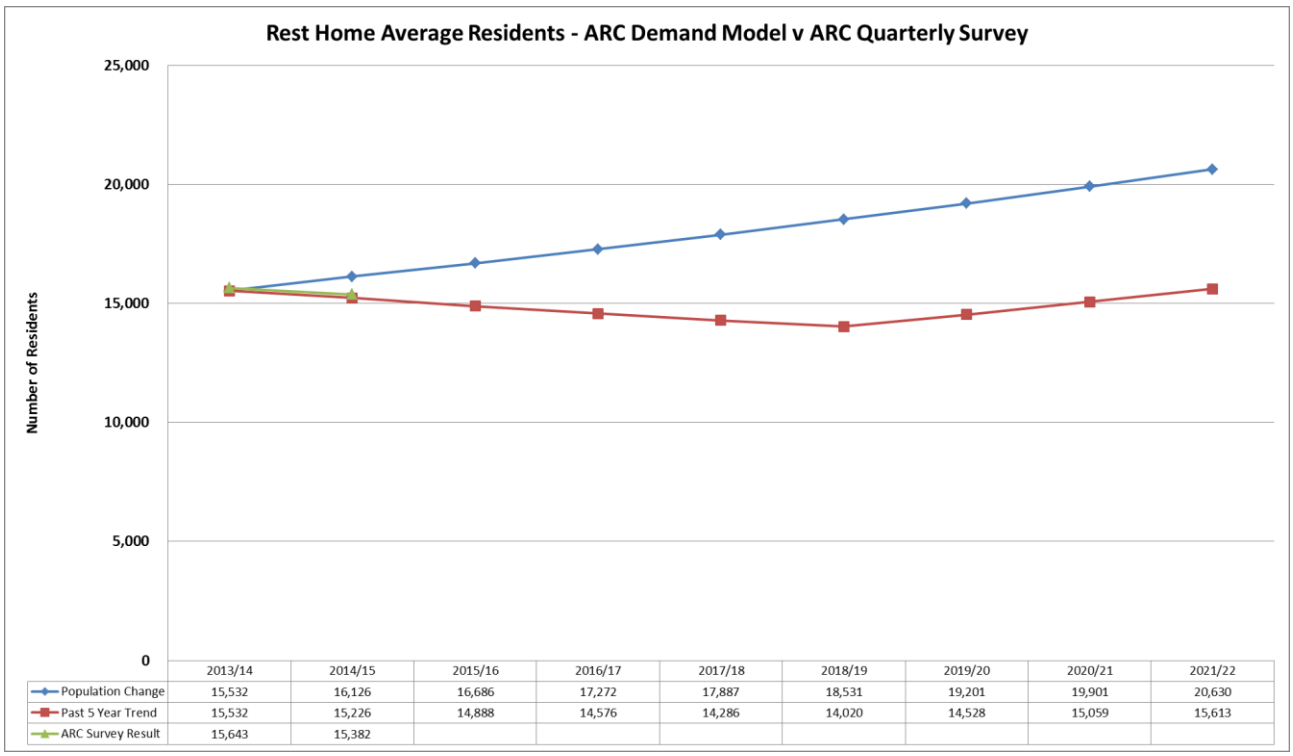
Rest Home Category

Figure 7: Rest Home Bed Totals - ARC Demand Model v ARC Quarterly Survey



	2013/14	2014/15	2015/16	2016/17	2017/18	2018/19	2019/20	2020/21	2021/22
Population Change	15,532	16,126	16,686	17,272	17,887	18,531	19,201	19,901	20,630
Past 5 Year Trend	15,532	15,226	14,888	14,576	14,286	14,020	14,528	15,059	15,613
ARC Survey Result	16,325	16,862							

Figure 8: Rest Home Average Residents - ARC Demand Model v ARC Quarterly Survey



	2013/14	2014/15	2015/16	2016/17	2017/18	2018/19	2019/20	2020/21	2021/22
Population Change	15,532	16,126	16,686	17,272	17,887	18,531	19,201	19,901	20,630
Past 5 Year Trend	15,532	15,226	14,888	14,576	14,286	14,020	14,528	15,059	15,613
ARC Survey Result	15,643	15,382							

Figure 7 above shows the projected Bed Days for the Population Chance and Past 5 Year Trend Scenarios, as well as the actual Bed Days from the ARC Survey.

Survey Results show a higher Beds supply than the Scenarios from the Demand Model, and results from the ARC Survey are closer to the Population Change than the Past 5 Year Trend Scenario.

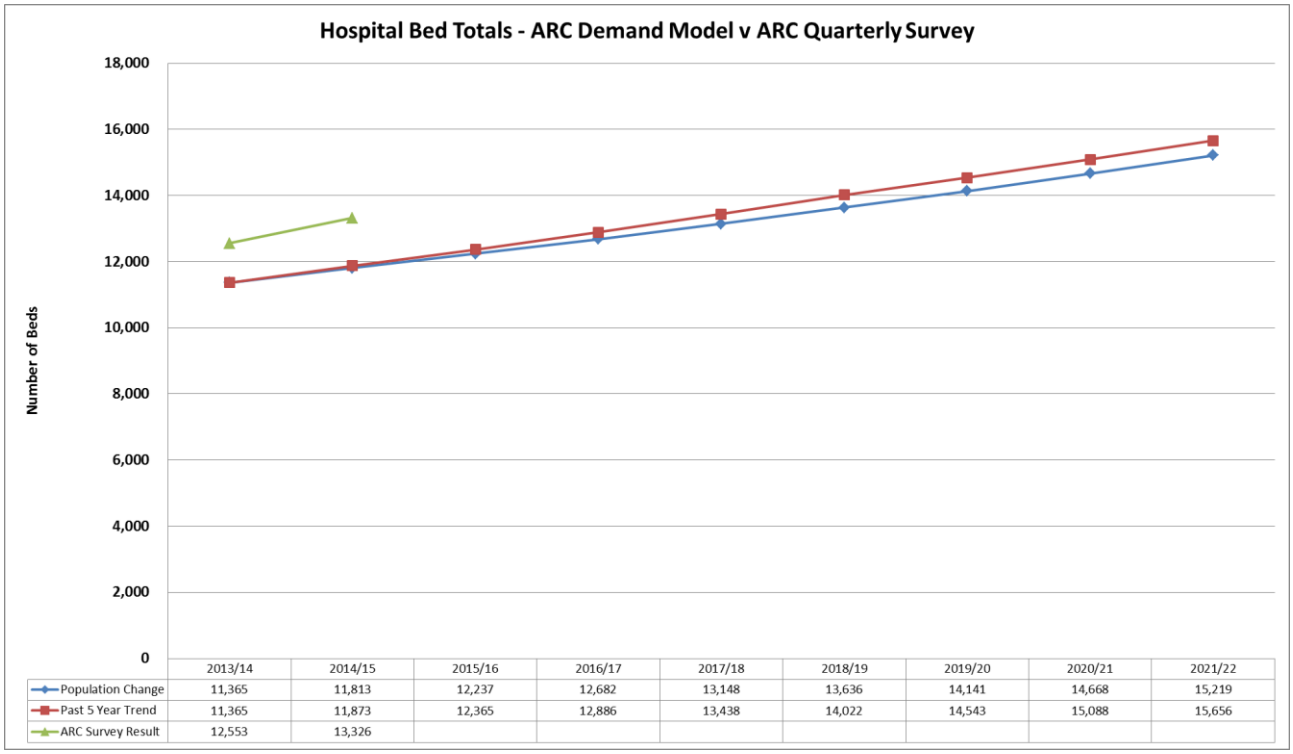
There is also a major difference between the two scenarios, as under the Past 5 Year Trend, an expected decrease over the next few financial years will occur, before projected demand for Bed Days increases back to 2013/14 Financial Year Levels.

Figure 8 shows similar results to Figure 7 regarding the two scenarios used in the Model. The chart indicates a decrease in the Average Residents per day that is closer to the Past 5 Year Trend. This is consistent with expectations that the number of Rest Home Residents will decrease over the next few years, while Hospital Residents will increase, however I will be tracking this more closely over the next few financial years.

Taken in concert, figures 7 and 8 lead to the conclusion of an oversupply of Rest Home Beds at a time when there is a decreasing number of Rest Home Residents.

Hospital Category

Figure 9: Hospital Bed Totals - ARC Demand Model v ARC Quarterly Survey



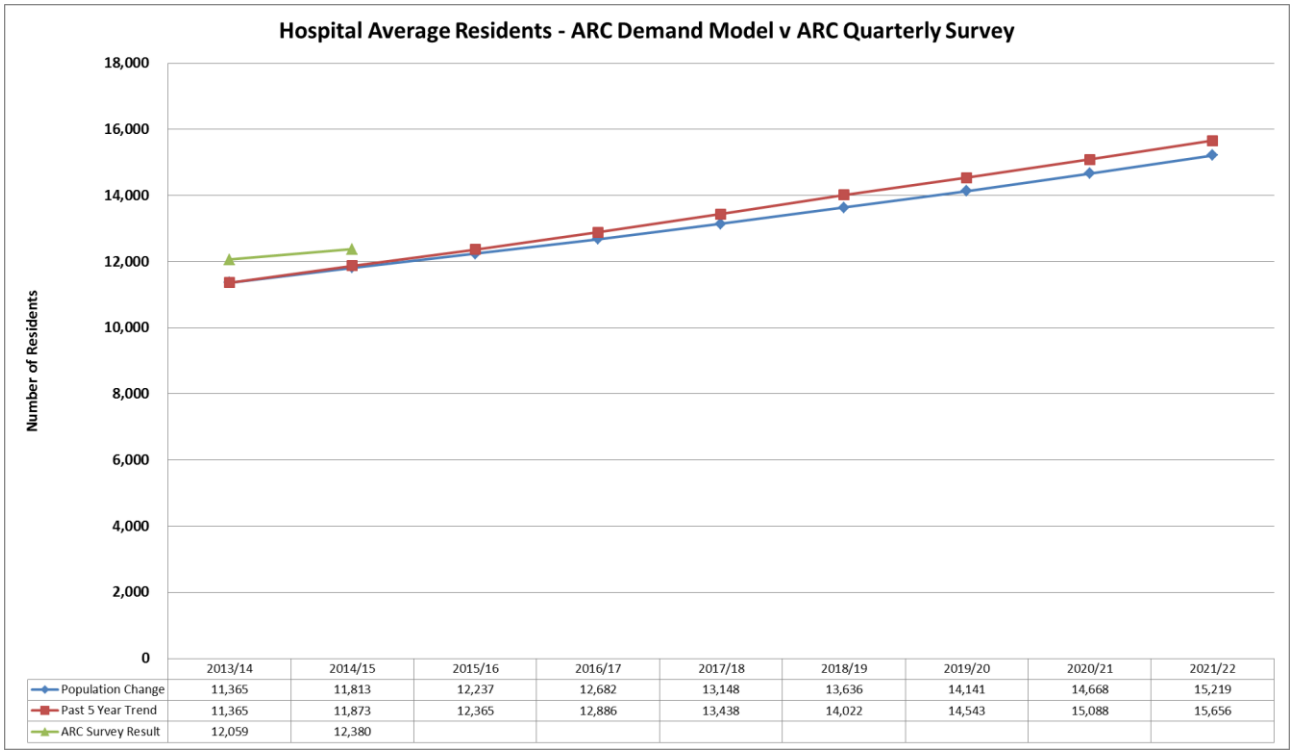
	2013/14	2014/15	2015/16	2016/17	2017/18	2018/19	2019/20	2020/21	2021/22
Population Change	11,365	11,813	12,237	12,682	13,148	13,636	14,141	14,668	15,219
Past 5 Year Trend	11,365	11,873	12,365	12,886	13,438	14,022	14,543	15,088	15,656
ARC Survey Result	12,553	13,326							

Figure 9 shows the projected Beds for the Population Chance and Past 5 Year Trend Scenarios, as well as the actual Beds from the ARC Survey. Survey Results show a higher Beds supply than the Scenarios from the Demand Model, with minor differences in outcomes between the two trends. The annual increase shown in the

Figure 10 shows the results for Average Residents/day from the Survey and the two Scenarios. Indications are that the Average Residents/day is at a lower rate that then expected supply shown in Figure 9.

Taken in concert, figures 9 and 10 show there to be a supply of Beds that is sufficient for the number of Average Residents/day.

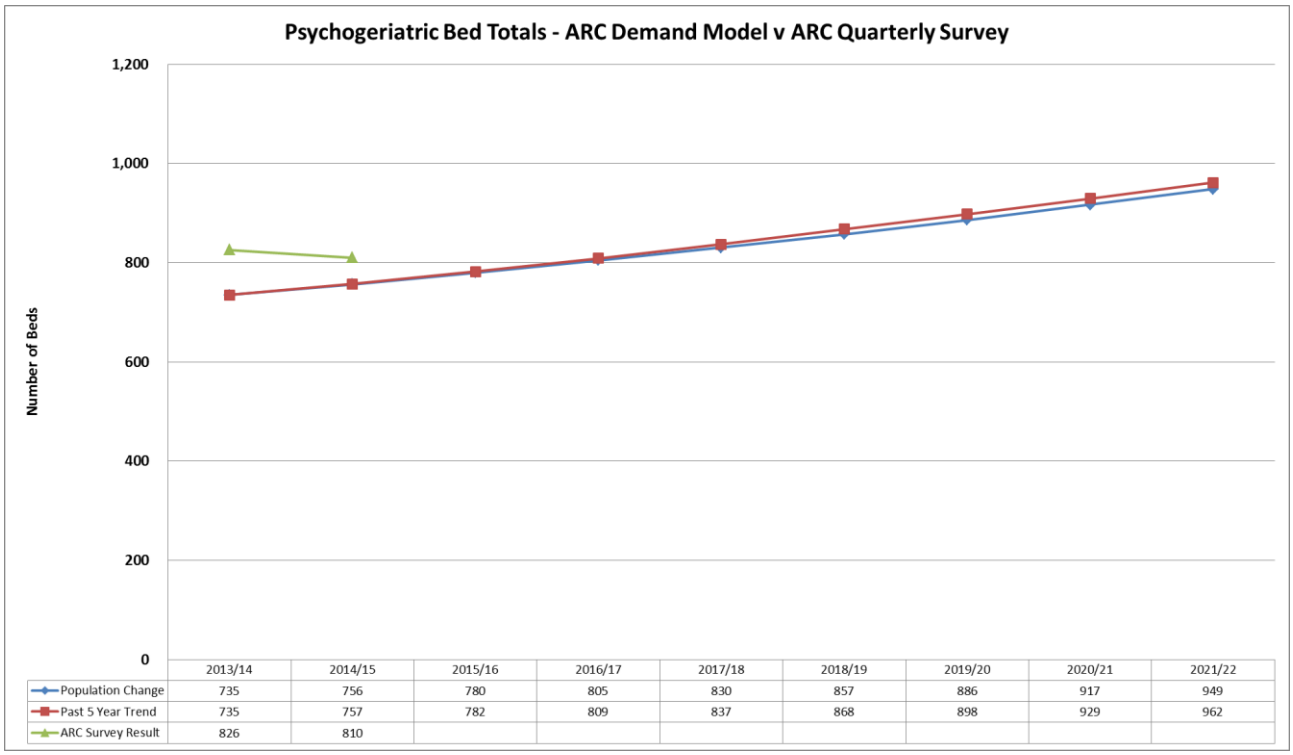
Figure 10: Hospital Average Resident Totals - ARC Demand Model v ARC Quarterly Survey



	2013/14	2014/15	2015/16	2016/17	2017/18	2018/19	2019/20	2020/21	2021/22
Population Change	11,365	11,813	12,237	12,682	13,148	13,636	14,141	14,668	15,219
Past 5 Year Trend	11,365	11,873	12,365	12,886	13,438	14,022	14,543	15,088	15,656
ARC Survey Result	12,059	12,380							

Psychogeriatric Category

Figure 11: Psychogeriatric Bed Totals - ARC Demand Model v ARC Quarterly Survey



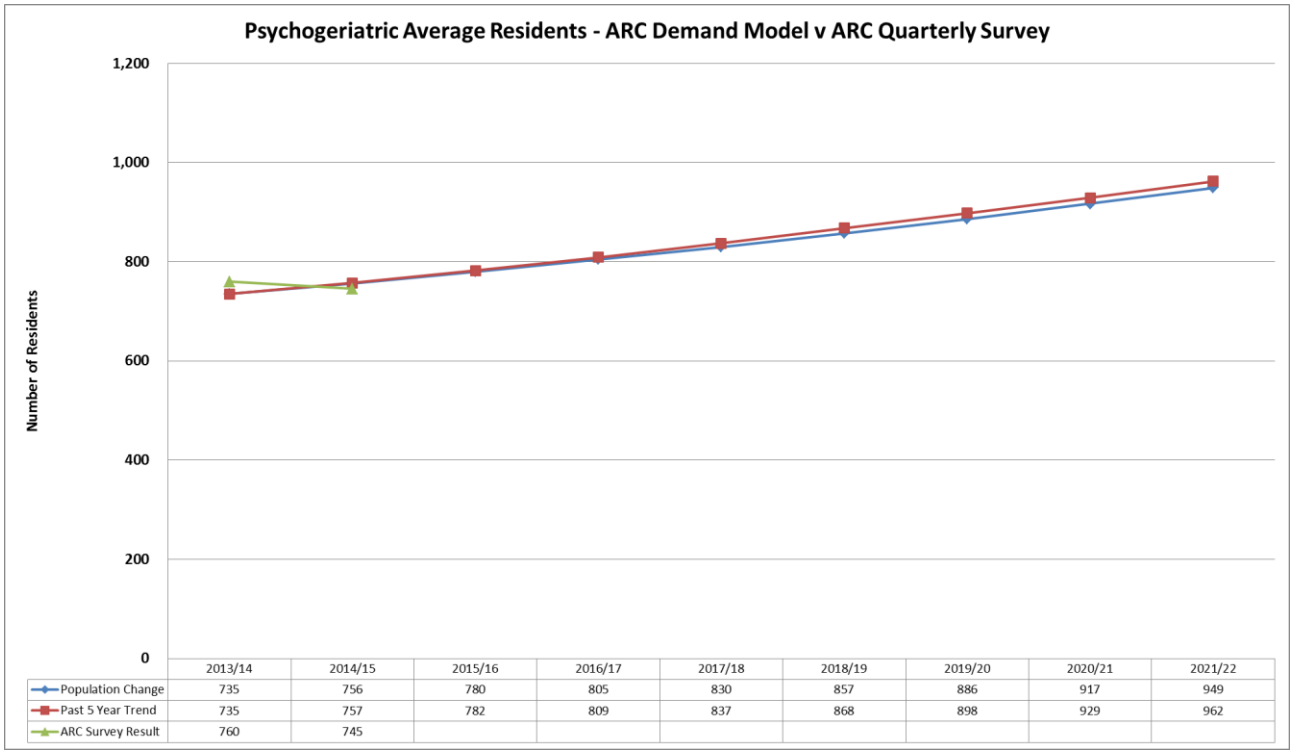
	2013/14	2014/15	2015/16	2016/17	2017/18	2018/19	2019/20	2020/21	2021/22
Population Change	735	756	780	805	830	857	886	917	949
Past 5 Year Trend	735	757	782	809	837	868	898	929	962
ARC Survey Result	826	810							

Figure 11 shows the Beds for the two scenarios tracking each other closely. More worrying is the decrease of Beds which is a contra trend, showing an actual decrease in the demand. This is also apparent in the

details in Figure 12, which show a decrease in the numbers of Average Residents/day, at a time when the forecast is showing an increase in the Average Residents/day.

Balancing these details, Beds available is still higher than Average Residents/day.

Figure 12: Psychogeriatric Average Residents Totals - ARC Demand Model v ARC Quarterly Survey



	2013/14	2014/15	2015/16	2016/17	2017/18	2018/19	2019/20	2020/21	2021/22
Population Change	735	756	780	805	830	857	886	917	949
Past 5 Year Trend	735	757	782	809	837	868	898	929	962
ARC Survey Result	760	745							

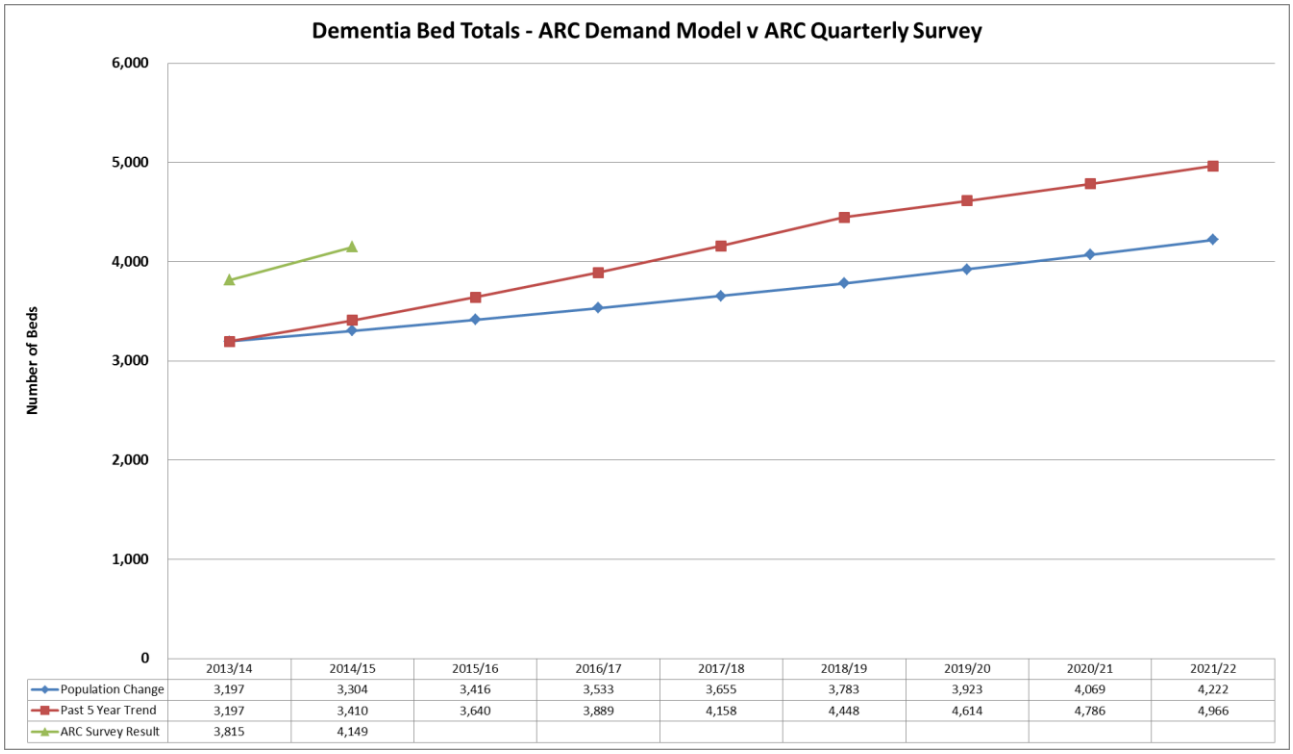
Dementia Category

Figure 13 indicates a higher correlation between the results of the ARC Survey for Rest Home Beds and the Population Change Trend than for the Survey result and the Past 5 Year Trend.

There is also a major difference with the Population Change Trend undergoing a gradual incline through the years while the Past 5 Year Trend shows a slightly higher incline to 2018/19 before easing off in growth.

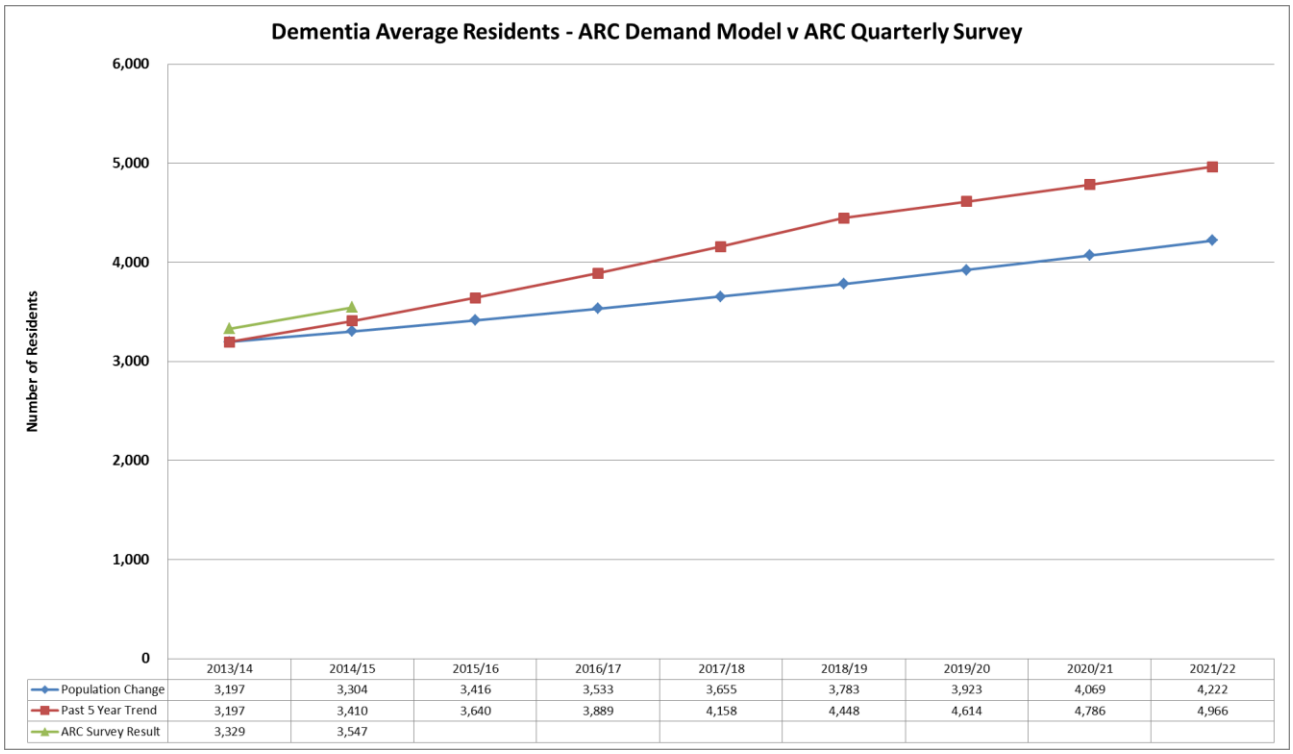
Taken in concert, figures 13 and 14 show there to be a supply of Beds that is sufficient for the number of Average Residents/day.

Figure 13: Dementia Bed Totals - ARC Demand Model v ARC Quarterly Survey



	2013/14	2014/15	2015/16	2016/17	2017/18	2018/19	2019/20	2020/21	2021/22
Population Change	3,197	3,304	3,416	3,533	3,655	3,783	3,923	4,069	4,222
Past 5 Year Trend	3,197	3,410	3,640	3,889	4,158	4,448	4,614	4,786	4,966
ARC Survey Result	3,815	4,149							

Figure 14: Dementia Average Residents - ARC Demand Model v ARC Quarterly Survey



	2013/14	2014/15	2015/16	2016/17	2017/18	2018/19	2019/20	2020/21	2021/22
Population Change	3,197	3,304	3,416	3,533	3,655	3,783	3,923	4,069	4,222
Past 5 Year Trend	3,197	3,410	3,640	3,889	4,158	4,448	4,614	4,786	4,966
ARC Survey Result	3,329	3,547							

Conclusion

The ARC Quarterly Survey Results shows a higher level of Available Beds than Average Residents/Day, however, with an insufficient number of data points for a trend line on the ARC Survey results, trends relating to the ARC Survey results may change. I will continue to monitor this and advise as necessary.